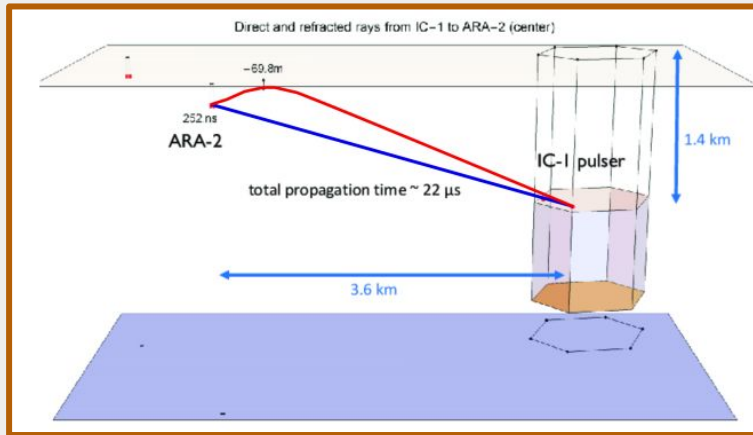


RNO-G Calibration Meeting

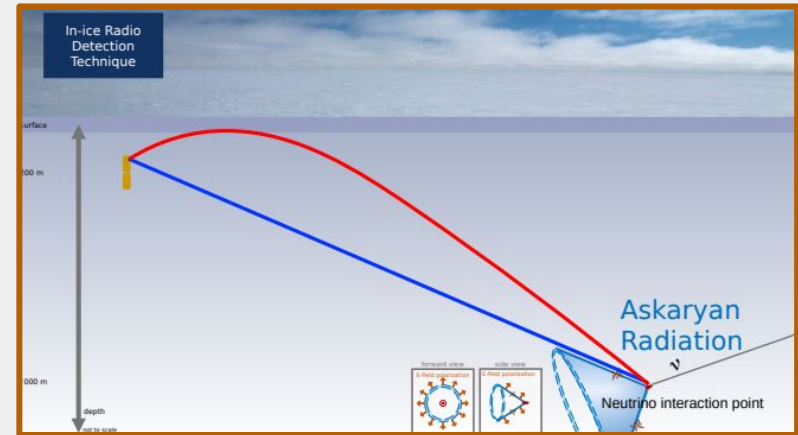
Direct and Refracted Signal Analysis at S21 from Saltzmann Hole

Motivation

- $\Delta t(D,R)$ is useful for determining shadow zone depths and understanding the refractive index model of ice
- Cherenkov cones can be modeled by calibration pulser signals [2]
 - Both tend to have a direct and secondary pulse at select geometries
- Amplitude ratios between D and R are useful for the investigation of ice properties within the ice surrounding stations



Cal pulser D and R raytracing [2]



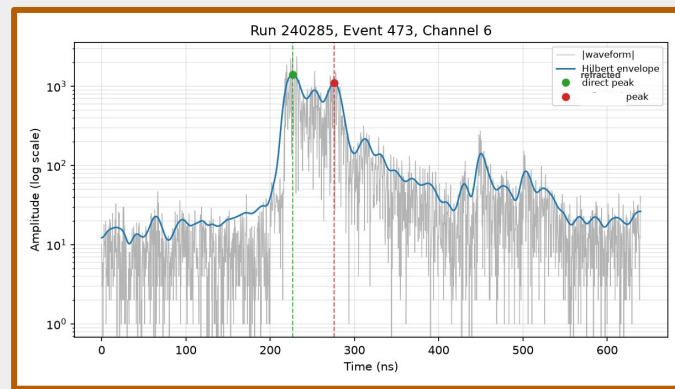
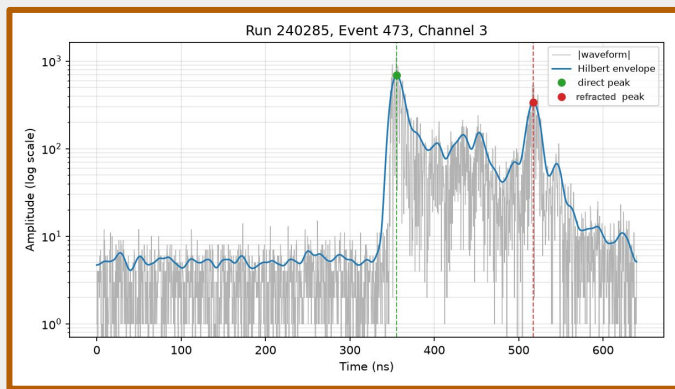
Neutrino event D and R raytracing [2]

Data

- Investigated ~10 cal pulser runs from August 1, 2024
 - Stepwise, taken from approx. 3pm-8pm local time
 - Each step was 5m
- GUNK antenna with IDL24 pulser inside, 2dB attenuation
 - Possible saturation of waveforms according to wiki, none I could see
- Best data was from 240m to 330m below the surface
- Only used Saltzmann hole to st21 Vpols (no Hpols)
- Distance from Saltzmann Hole is ~680m from st21
- August 6, 2024 (20dB attenuation) signal was not strong enough for our current pipeline, more work might be done with that data in the future

Common Methods (Needed for both Amp. Ratio and $\Delta t(D,R)$)

- Used waveforms from runs 240283-240293 from station 21
- Took peak-to-peak size and divided it by $2 \times \text{RMS}$ to get SNR
 - Anything with an SNR above 13 was passed, all else was removed from sample
- Used hilbert envelope with a very smooth gaussian filter (13 sigma)
- Found 2 highest peaks (needed to be at least 30ns apart), identified the first as D and the second as R
 - Recorded time and amplitude values for each peak in each channel
 - Done for every event with $\text{SNR} > 13$



D and R Time Delay Methods

- Only interested in channels 0-3, 5, 6, 22, and 23 (Vpols)
 - Did not use 9 or 10 due to a lack of simulated 3-stage model time delay values
- Plotted time difference by channel of D and R signal peaks as a function of the transmitter depth in which they occur

D and R Amplitude Ratio Methods

- Still interested only in select channels above
- Found the amplitude ratio of each D and R pair by dividing the channel R/D amplitudes
- Plotted these ratios by channel as a function of transmitter depth

D and R Ray Tracing and Amplitude Simulation

- Used NuradioMC main simulation located in GitHub
- Created a config file and used the detector station .json file to feed into the simulation
 - Chose pulser signal model, greenland simple ice model, analytic raytracing, auto polarization
 - Focusing set to True, no reflections allowed, and noise set to false
- Recorded delay times for direct and refracted straight from simulation
- Originally used Friis equation to simulate amp. ratio decay
 - Then added a cosine-like gain pattern to simulate the amplitude ratio oscillations
 - Eventually used a linear equation with cosine-like gain pattern due to the misfit of the Friis equation with our data

$$\text{ratio}_{\text{Friis}}(z) = \frac{A_R(z)}{A_D(z)} = \frac{\left(\frac{|\cos \theta_{Tx,R}| |\cos \theta_{Rx,R}|}{r_R} \right)}{\left(\frac{|\cos \theta_{Tx,D}| |\cos \theta_{Rx,D}|}{r_D} \right)}$$

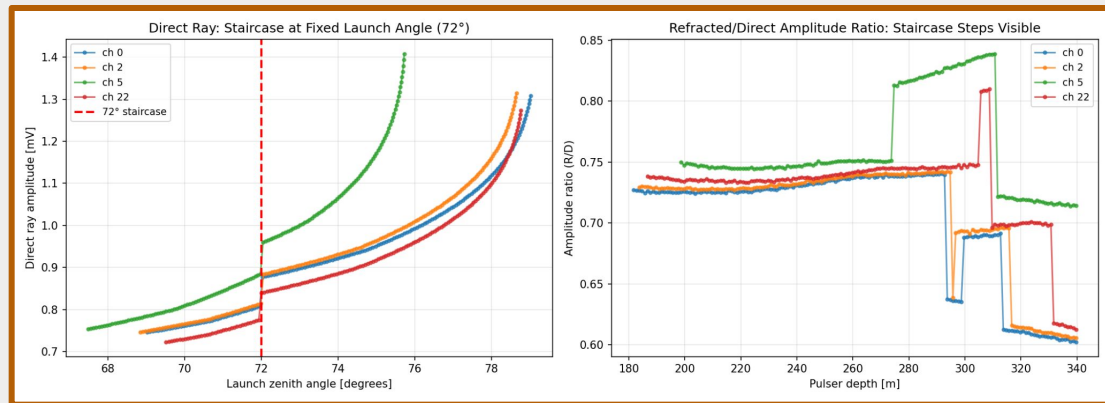
$$\text{ratio}_{\text{measured}}(z) \approx \text{ratio}_{\text{Friis}}(z) \times \cos\left(\frac{2\pi z}{\lambda} + \varphi_0\right)$$

Why Not Amplitudes from NuRadioMC Outputs?

- Decided to go with Friis equation fits and linear fits with cosine-like gain because NuRadioMC has massive jumps in direct amplitudes around 72 and 54 degree launch angles
 - This affects the amplitude ratios, causing massive discontinuities
- Investigated multiple theories, including the emitter/antenna files and seeing if they had these jumps (they didn't)

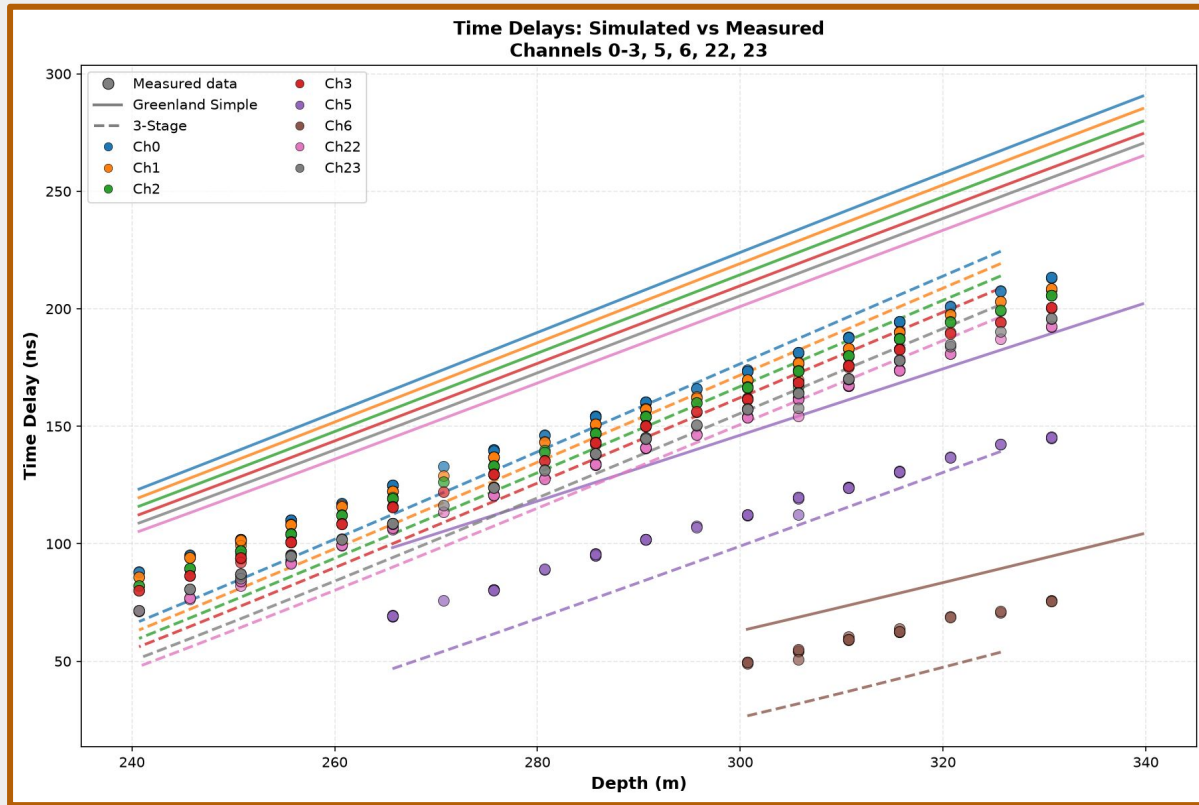
Simulation emitter and receiver models:

- rnog_cal5c_0db.hdf5 in the pulser_waveforms folder in NuRadioMC
- RNOG_vpol_4inch_center_n1.73 folder in AntennaModels in NuRadioReco



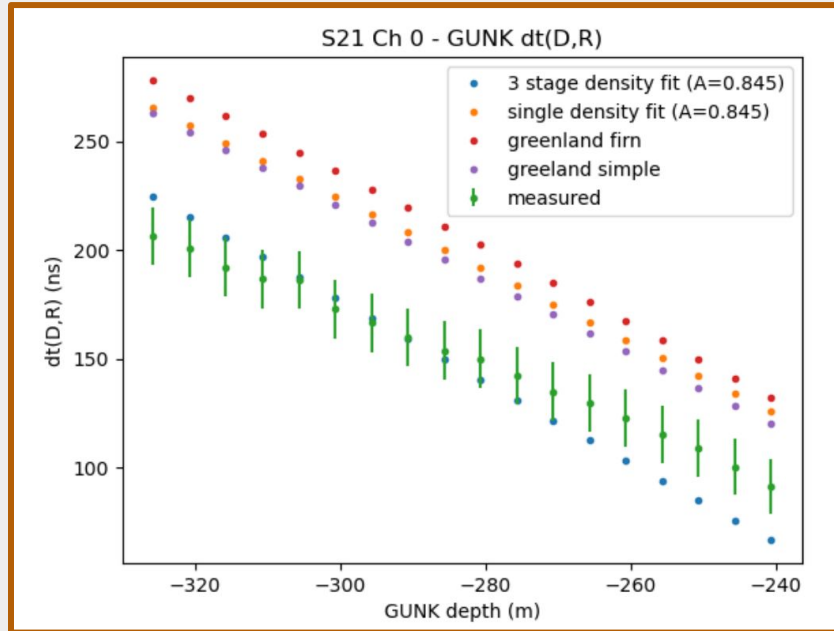
(left) direct amplitudes as a function of launch angle, (right) amplitude ratios as a function of depth. Both from NuRadioMC

D and R Time Delay Results

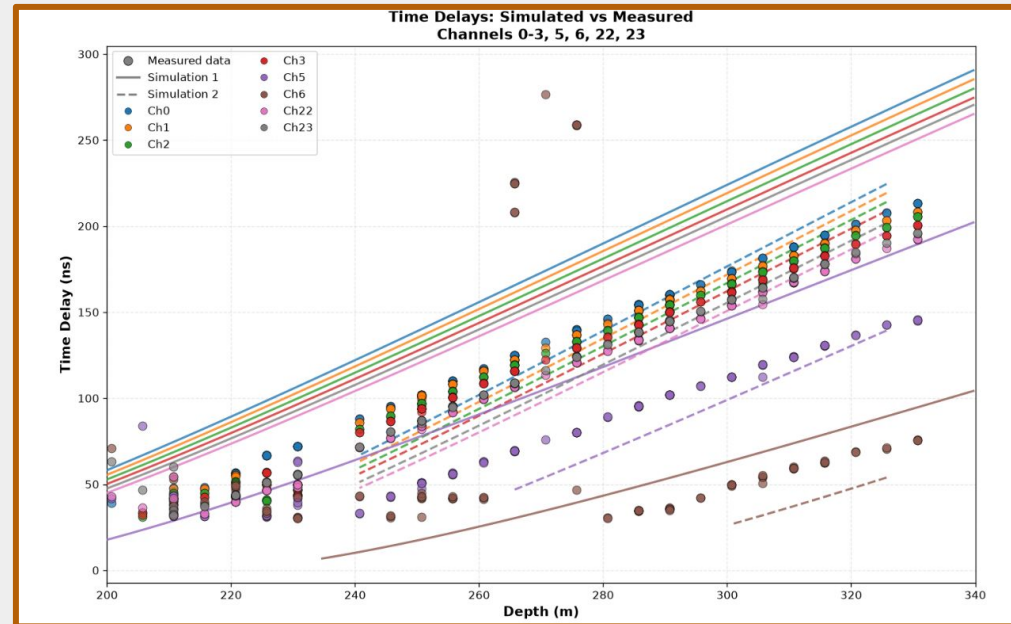


Time delay between direct and refracted signals during August 1, 2024 calibration pulser runs - Saltzmann Hole

D and R Time Delay Results

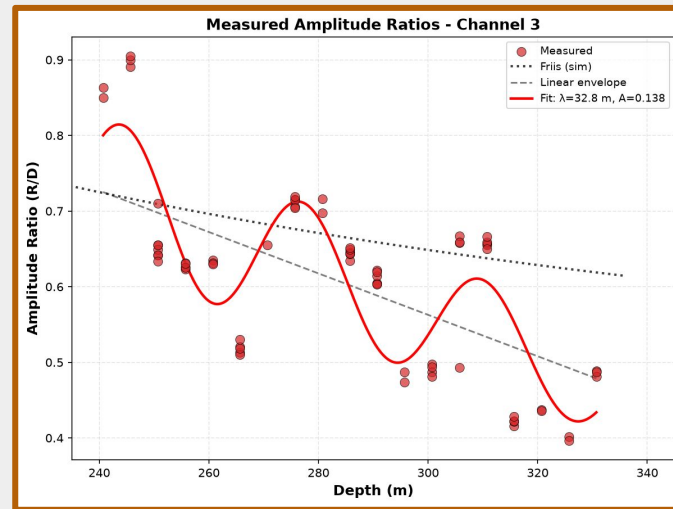
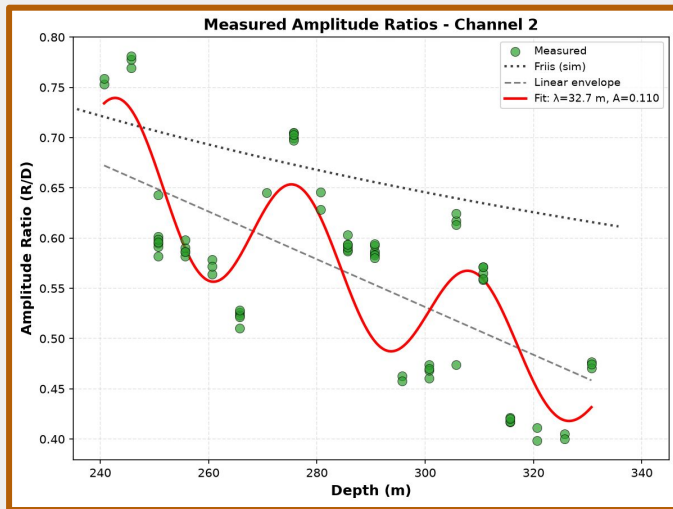
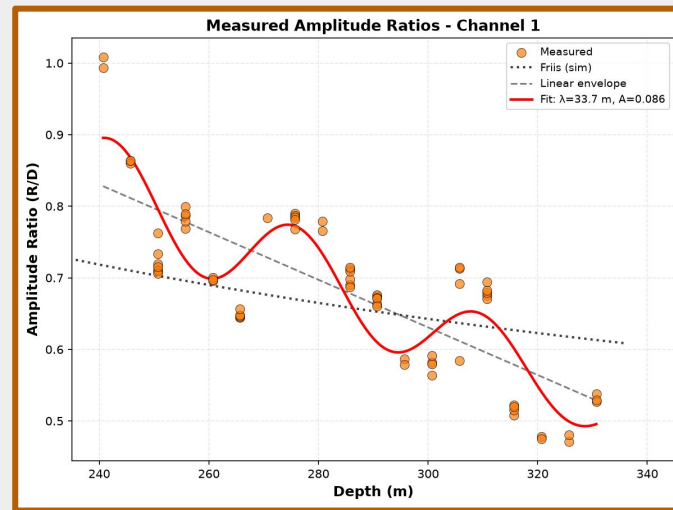
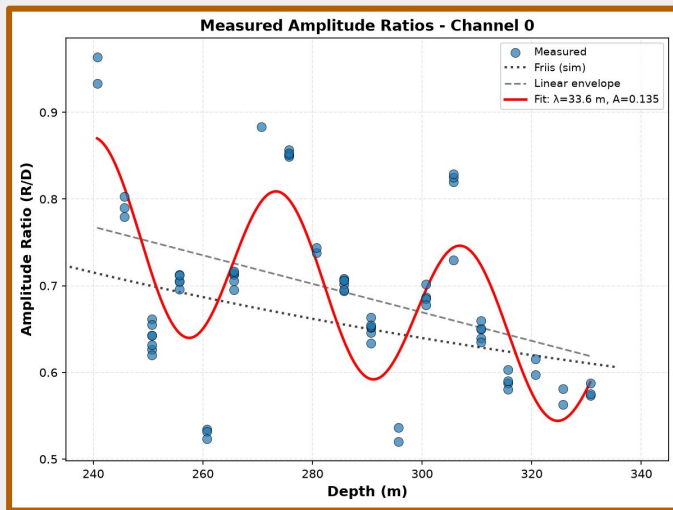


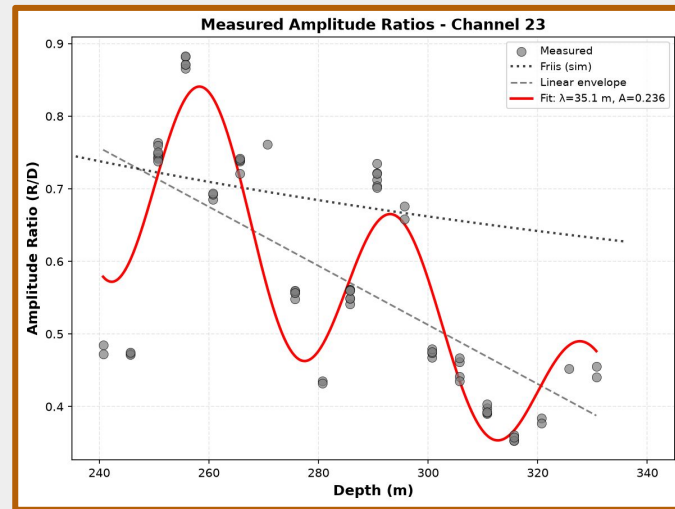
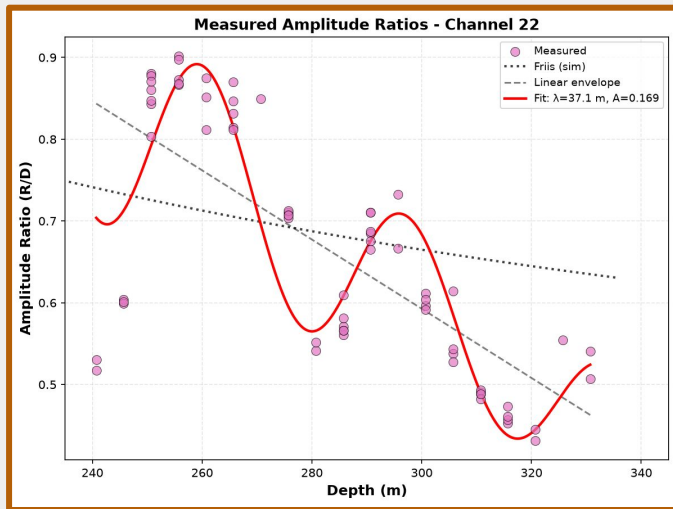
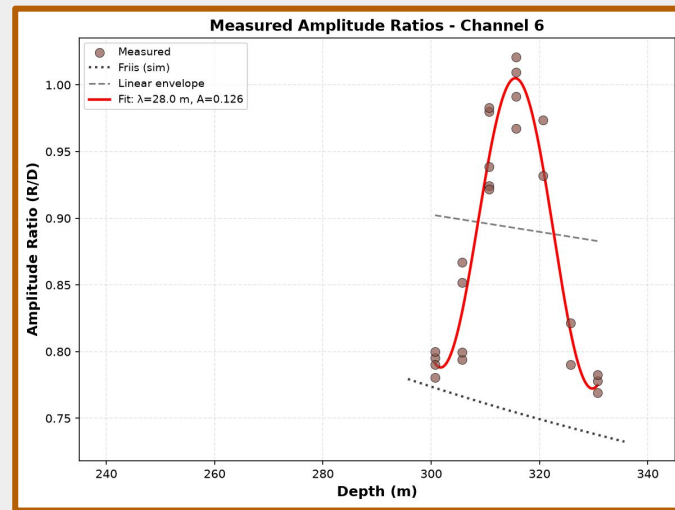
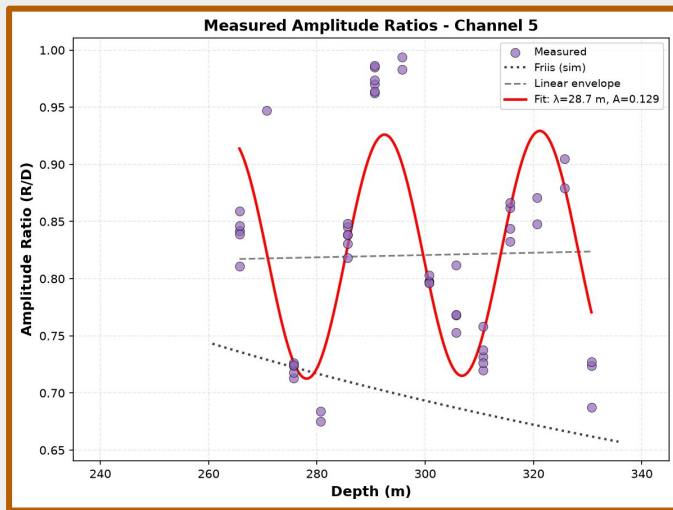
Previous RNO-G August 1, 2024 Saltzmann
Hole time delay analysis featuring multiple
model comparisons



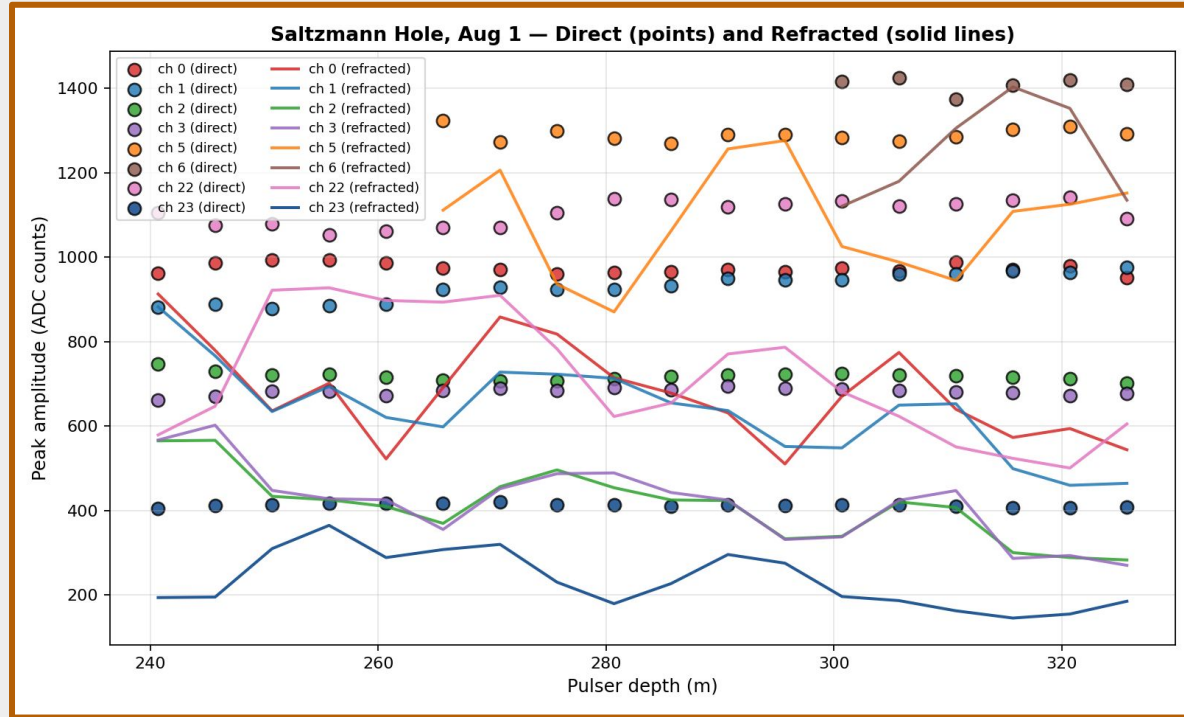
My time delays including Tx signals past the
caustic for each channel

D and R Amplitude Ratio Results





Which Signal was Actually Oscillating with Depth? D or R? (It's the Refracted...)



Direct and refracted signal amplitudes plotted on top of one another for August 1, 2024 Saltzmann Hole pulser runs

Additional Simulation Specifications and Methods

- Methods for amplitude ratio linear fitting with cosine-like gain patterns:
 - Np.polyfit data to fit the linear curve to the data
 - Fit 400 wavelength values between 15 and 60 with 8 possible starting phase values, chose best combination
 - More iterations around best combination to minimize error
 - Amplitude solved in this way too
- Statistic results for amp. ratio fits:

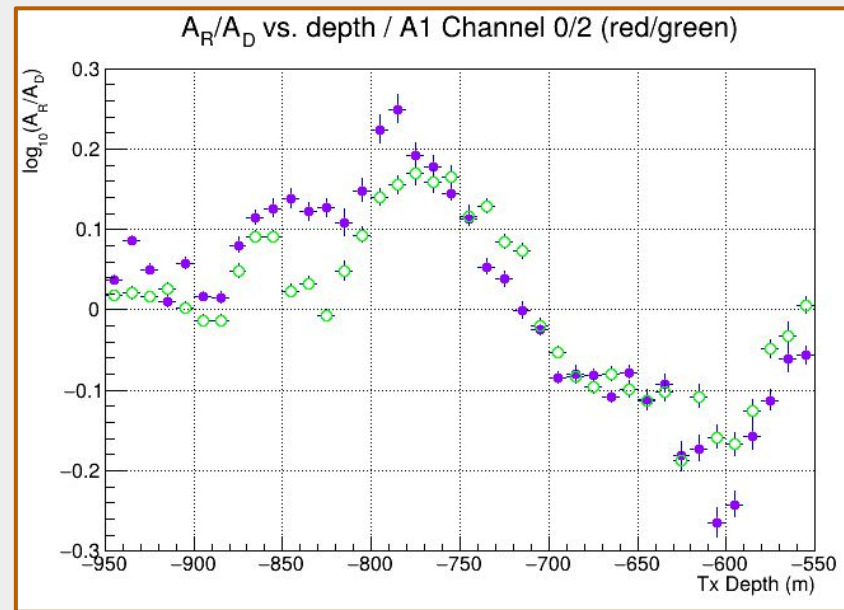
Ch	λ (m)	ϕ_0 (rad)	A_osc	slope m	b	χ^2_{red}	env-sim	N
0	33.59±0.82	-0.970	0.1349	-1.645e-03	1.1629	0.00419	+0.0362	71
1	33.71±1.26	-1.189	0.0856	-3.327e-03	1.6287	0.00324	+0.0218	71
2	32.69±1.02	-2.858	0.1099	-2.376e-03	1.2441	0.00311	-0.0959	71
3	32.79±0.98	-2.859	0.1375	-2.738e-03	1.3841	0.00483	-0.0618	71
5	28.68±0.77	-1.232	0.1294	1.004e-04	0.7905	0.00272	+0.1217	51
6	28.02±1.36	-1.674	0.1259	-6.443e-04	1.0960	0.00063	+0.1368	24
22	37.12±1.01	-0.084	0.1685	-4.231e-03	1.8619	0.00446	-0.0200	69
23	35.05±0.77	-2.485	0.2360	-4.069e-03	1.7331	0.00472	-0.0999	69

Notable Mentions and Odd Behavior

- Ch. 23 consistently shows much lower amplitudes than all other channels (even 22)
- Ch. 7 (most shallow detector) is not useful for any of the Saltzmann hole calibration data due to signal arrive outside the waveform capture
 - Even the lowest depth of 330m below surface
- Ch. 22 and 23 waveform refracted signal was cut off at depths exceeding 325m
 - Due to limited sample size of waveform
- Hpol shows intense amplitudes (almost as much as Vpol channels)
 - Strange considering GUNK is a vpol antenna
 - Hpol amplitude ratios seem to be higher than the Vpol amplitude ratios on August 1, 2024

Take Away and Conclusions

- RNO-G amplitude ratios show same patterns seen in south pole ice with ARA
- Time delay measurements match previous analysis (by Kenny)
 - 3-stage model still seems to be superior
 - Shallower channels need a better model - neither were close
- Shadow zones for shallower Vpols (5 & 6) prevent refracted signals at certain transmitter depths



ARA log amplitude ratio between R and D as a function of depth

References

- [1] Ali, S., Allison, P., & Archambault, S., et al. (2026). *Modeling the refractive index profile $n(z)$ of polar ice for ultra-high energy neutrino experiments*. Arxiv.Org. <https://arxiv.org/html/2406.00857v4>
- [2] Allison, P., Archambault, S., J.J. Beatty, & D.Z. Besson, et al. (2020). Long-baseline horizontal radio-frequency transmission through polar ice. *Journal of Cosmology and Astroparticle Physics*, 2020(12), 009–009. <https://doi.org/10.1088/1475-7516/2020/12/009>